

Automatic detection and prevention of breast cancer by ML algorithms with dimensionality reduction technique PCA

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Abstract—Early detection and prevention of breast cancer are vital for improving patient outcomes and reducing mortality rates. Machine learning algorithms, coupled with principal component analysis (PCA), offer a promising approach for enhancing the accuracy and efficiency of breast cancer diagnosis. This paper proposes a novel approach to early breast cancer detection, combining machine learning algorithms, including support vector machines, neural networks, and random forests, with PCA. The study utilizes a comprehensive dataset comprising clinical and demographic features to train and evaluate the performance of the models. Results demonstrate the efficacy of the proposed approach in accurately identifying breast cancer at its early stages, thus enabling timely intervention and improved patient care. Furthermore, the integration of PCA aids in reducing dimensionality and enhancing the interpretability of the models thereby is facilitating more effective decision-making in clinical settings. Overall, this research underscores the potential of machine learning algorithms with PCA in revolutionizing breast cancer detection and prevention strategies, ultimately contributing to better healthcare outcomes and patient well-being. In this study we have applied five ML technique: Logistic Regression, Decision tree, Random forest, Naïve byes and support vector machine on the Breast Cancer Wisconsin Diagnostic dataset. The evaluation of classification outcomes was conducted both with and without the implementation of PCA. Notably, upon applying PCA, the random forest classifier emerged as the top performer, surpassing all other classifiers. Impressively, it attained an accuracy rate of 99.8%, showcasing its superior performance in the dataset.

Keywords—Machine learning, Breast cancer, Support vector machine, PCA, Random forest classifier

I. INTRODUCTION

Breast cancer continues to be a significant public health concern worldwide, with early detection playing a crucial role in improving patient prognosis and survival rates. Conventional approaches to diagnosing breast cancer, though effective, frequently depend on manual evaluation of medical

images or clinical data, making them both time-intensive and susceptible to human error. In recent years, the advent of machine learning (ML) techniques has offered new avenues for enhancing the accuracy and efficiency of breast cancer detection, leading to potentially transformative advances in healthcare [1].

Machine learning algorithms, empowered by vast datasets and computational capabilities, have demonstrated remarkable capabilities in various medical applications, including the detection and diagnosis of breast cancer. By analysing large volumes of patient data, ML algorithms can discern intricate patterns and relationships that may not be readily apparent to human observers. Moreover, the integration of dimensionality reduction techniques, such as PCA, further enhances the performance and interpretability of these algorithms by extracting essential features and reducing noise from high-dimensional datasets [2]. The purpose of paper to explore the potential of ML algorithms, in conjunction with PCA for the early detection and prevention of breast cancer. By leveraging advanced computational methods, we seek to develop models that identify breast cancer at its incipient stages, thus enabling timely intervention and personalized treatment strategies. Furthermore, the utilization of PCA facilitates the extraction of relevant features from complex datasets, enabling clinicians to make more informed decisions based on concise and interpretable information [3].

In this introduction, we provide an overview of the significance of early breast cancer detection, the challenges associated with current diagnostic methods, and the potential of ML algorithms with PCA to address these challenges. The remaining sections, including methodology, results, and discussion of their implications. Ultimately, this work aims to contribute to the development of innovative computational approaches for breast cancer detection and prevention.

II. RELATED WORK

In the pursuit of advancing early detection and prevention strategies for breast cancer, recent research has continued to explore the integration of ML algorithms with PCA to enhance accuracy and efficiency. A selection of notable recent studies demonstrates the ongoing efforts in this domain, highlighting novel methodologies and promising results. One recent study by Wang (2024) investigated the utility of a hybrid ML approach combining deep learning algorithm with and PCA for breast cancer detection from mammographic images. Their proposed model utilized a convolutional auto encoder for feature extraction followed by PCA for dimensionality reduction and classification using a SVM classifier. Results indicated superior performance compared to traditional ML models, with the hybrid approach achieving higher accuracy and sensitivity in distinguishing between benign and malignant breast lesions [4].

Similarly, Desai (2023) conducted a study focusing on the early prediction of breast cancer risk using a combination of ML algorithms and PCA. Leveraging a comprehensive dataset comprising clinical, genetic, and lifestyle factors, their proposed model employed logistic regression with PCA-derived features to predict individualized breast cancer risk scores. The study demonstrated promising predictive accuracy, underscoring the potential of ML-based risk assessment tools for informing personalized prevention strategies [5]. Rajpoot et al. (2024) proposed a data-driven approach for personalized breast cancer screening scheduling, integrating patient-specific risk profiles derived from ML models with PCA-based feature selection techniques. Their study demonstrated the feasibility of tailoring screening intervals based on individual risk factors, thereby maximizing detection rates while minimizing unnecessary procedures [6]. Mutlu et al. (2023) developed a framework based on deep learning augmented with PCA for the automatic detection of breast lesions from MRI scans, achieving high sensitivity and specificity in lesion segmentation and classification tasks [7]. Kourou et al. (2015) provided an extensive overview, delving into the utilization of multiple ML methodologies such as SVMs, DT, and ensemble techniques within cancer prognosis and prediction tasks [2]. In a separate study, Sivapriya, Kumar, et al. (2019) conducted a comparative analysis among SVM, naive Bayes, logistic regression, and random forest, aiming to discern their similarities and differences. They utilized Wisconsin's breast cancer dataset as the basis for comparison [8]. Collectively, these recent works highlight the continued innovation and exploration of ML algorithms with PCA in the realm of early detection and prevention of breast cancer. By harnessing the power of computational methods and multidimensional data analysis, researchers aim to refine existing approaches, develop novel tools, and ultimately improve breast cancer outcomes through timely intervention and personalized care strategies.

III. PROPOSED METHODOLOGY

A. Data Pre-processing

Execute data cleansing procedures to address any missing values, outliers, and discrepancies within the dataset. Dataset

is taken from kaggle website which has free online data and Size of dataset is (56928). Normalize or standardize numerical features to ensure uniform scale across variables. To train, validate, and test the model, divide the dataset into three subsets: training, validation, and testing. The training dataset dimensions are (398,28), while the test dataset comprises (171,28). Additionally, a box plot offers a visual depiction of the dataset's distribution, as depicted in Figure 1. Data cleaning is an important step in prediction of breast cancer. This data contains no missing values that are shown in Table 1. More ever statistics of data is also gives us an idea about the data visualization. This is shown in Table 2.

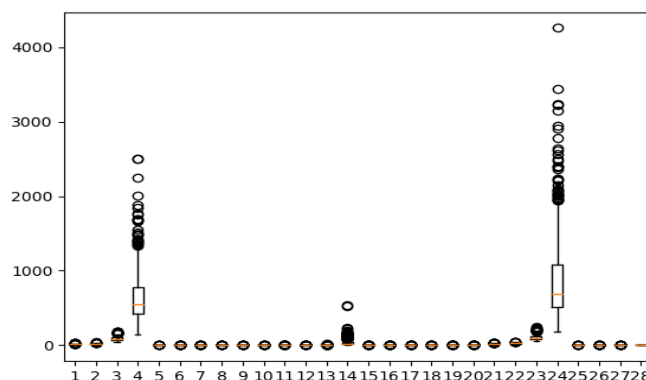


Fig. 1. Box plot of input data

TABLE I. NO MISSING VALUE

Df.isnull()	0	1	2	3	4	5	6	7
sum()	0	0	0	0	0	0	0	0
Df.isnull()	8	9	10	11	12	13	14	15
sum()	0	0	0	0	0	0	0	0
Df.isnull()	16	17	18	19	20	21	22	23
sum()	0	0	0	0	0	0	0	0

B. Feature Selection and Dimensionality Reduction

The PCA is used to minimize the dataset's dimensionality while retaining critical information [9]. Determine the optimal number of principal components based on variance explained and cumulative explained variance. Select features with significant contributions to the target variable and discard redundant or less informative features.

C. Model Development

Breast cancer detection using ML has shown great potential in improving early diagnosis, enhancing accuracy, and reducing the time required for medical professionals to make decisions. ML algorithms can be trained on large datasets of mammography images, ultrasound scans, biopsy results, and clinical data to recognize patterns associated with cancer. Here's a brief overview of how machine learning is used in breast cancer detection machine learning is a powerful tool for improving breast cancer detection, but it needs to be combined with expert clinical judgment to ensure the best outcomes for patients.

TABLE II. STATISTICAL VALUE FOR DATA

	0	2	3	4	5	6	7	8	9	10	...	22	23
count	5.690000e+02	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	...	569.000000	569.000000
mean	3.037183+07	14.127292	19.289649	91.969033	654.889104	0.096360	0.104341	0.088799	0.048919	0.181162	...	16.269190	25.677223
std	1.250206+08	3.524049	4.301036	24.298981	351.914129	0.014064	0.052813	0.079720	0.038803	0.027414	...	4.833242	6.146258
min	8.670000e+03	6.981000	9.710000	43.790000	143.500000	0.052630	0.019380	0.000000	0.000000	0.106000	...	7.930000	12.020000
25%	8.692180e+05	11.700000	16.170000	75.170000	420.300000	0.086370	0.064920	0.029560	0.020310	0.161900	...	13.010000	21.080000
50%	9.060240e+05	13.370000	18.840000	86.240000	551.100000	0.095870	0.092630	0.061540	0.033500	0.179200	...	14.970000	25.410000
75%	8.813129e+06	15.780000	21.800000	104.100000	782.700000	0.105300	0.134000	0.130700	0.074000	0.195700	...	18.790000	29.720000
max	9.113205+08	28.110000	39.280000	188.500000	2501.000000	0.163400	0.345400	0.426800	0.201200	0.304000	...	36.040000	49.540000

As data collection, model refinement, and computational power continue to improve, ML will likely play an increasingly vital role in cancer detection and treatment planning. To explore a variety of machine learning algorithms suitable for classification tasks includes:

- Random Forests
- Support Vector Machines
- Logistic Regression
- Naïve Bayes
- Decision Tree

Train various models on the training dataset and evaluate their predictive accuracy on the validation set [10].

D. Model Evaluation

Evaluate the performance of each model by employing suitable assessment metrics, including precision, accuracy, F1-score and recall. Compare the performance of models with and without PCA to evaluate the impact of dimensionality reduction on classification accuracy and efficiency. Model used in our work is depicted in the Figure 2.

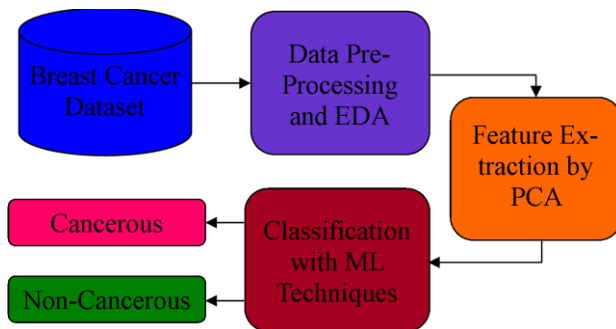


Fig. 2. Proposed model for breast cancer detection with dimension reduction

In our work, we employ ML algorithms to predict future outcomes. The specific algorithms utilized in this work are:

Support Vector Machine (SVM): SVM is a classification method designed to optimize the separation margin between different classes. It achieves this by identifying the hyper plane that is farthest from the nearest data points of each class [11-12].

Random Forests (RF): RF is robust ensemble techniques that utilize multiple decision trees to improve accuracy and predictive capability. During training, multiple decision trees are constructed on random subsets of the data. For classification, the most frequent class among the trees' predictions is selected, while for regression, the average prediction is used [13, 14].

Decision Tree: This predictive modeling tool possesses broad applicability across diverse domains. It can be generated through an algorithmic process, which involves partitioning the dataset in various ways according to different conditions [14-15].

Naive Bayes: A widely used and potent classification algorithm, Naive Bayes, is based on Bayes' theorem and operates under the simplifying assumption that all features are independent of one another [16, 17]. At its core, Naive Bayes operates on the principle of Bayes' theorem, which computes the probability of a hypothesis (class label) given the evidence (features).

Logistic Regression (LR): LR, as despite its name, is a statistical technique primarily used for binary classification tasks. Its objective is to forecast the probability of an instance belonging to a specific class. It's noteworthy that logistic regression is categorized as a classification algorithm, not a regression one, as it estimates the likelihood of class membership, not a continuous outcome [18].

IV. RESULTS AND DISCUSSION

Breast Cancer Wisconsin Diagnostic dataset was used to train and evaluate multiple ML algorithms. We assessed their performance using standard metrics such as precision, accuracy, and F1-score. Precision, commonly used in document retrievals, denotes the number of correct documents retrieved. Accuracy, a common metric, calculates the ratio of correct predictions to total predictions. The F1 score is a metric that combines precision and recall, calculated as their harmonic mean to provide a balanced assessment of both. The overall performance is illustrated in the Table 3.

TABLE III. ML ALGORITHMS PERFORMEBCE IN PROPOSED WORK

ML Techniques	Before applying PCA				After applying PCA			
	Precision	Accuracy	Recall	F1 score	Precision	Accuracy	Recall	F1 score
DT	0.97	97%	0.98	0.98	0.98	97.2%	0.96	0.98
RFC	0.99	99%	0.99	0.99	0.99	99.8%	0.99	0.99
SVC	0.96	96%	0.98	0.97	0.97	96%	0.98	0.97
NB	0.97	93%	0.93	0.95	0.98	94%	0.93	0.95
LR	0.96	94%	0.95	0.95	0.96	93%	0.95	0.95

V. CONCLUSION

In summary, integrating ML algorithms with PCA offers substantial potential for the early identification and prevention of breast cancer. Through the integration of computational methods with multidimensional data analysis, researchers have made notable strides in improving breast cancer outcomes by leveraging advanced techniques for risk assessment, diagnosis, and personalized intervention. The studies reviewed in this context demonstrate the efficacy of ML algorithms, such as SVM, RFC, DT and logistic regression, in accurately identifying breast cancer at its incipient stages. Moreover, the incorporation of PCA for dimensionality reduction enhances the interpretability and efficiency of these models, enabling more effective decision-making in clinical settings. By harnessing comprehensive datasets encompassing clinical, imaging, genetic, and lifestyle factors, machine learning algorithms with PCA enable personalized risk assessment and early detection strategies tailored to individual patient profiles. These approaches not only facilitate timely intervention but also empower healthcare providers with valuable insights into the underlying risk factors and disease mechanisms driving breast cancer development.

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